25900448

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEHSC202 – SPORTS NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **2** marks

1. Define aerobic capacity. [K]/ [CO1]
   - Refers to the maximum amount of oxygen the body can utilize during intense or prolonged exercise**.** (2 marks)

2. Mention two factors affecting fluid requirement during exercise. [U] / [CO2]
   - Intensity of Exercise (1 mark)
   - Environmental Conditions (1 mark)

3. Give 2 examples of protein rich foods suitable for athletes. [U] / [CO3]
   - Animal based sources: Meat/ egg (1mark)
   - Plant based source: Whole grains/legumes (1 mark)

4. State 2 physical benefits of regular exercise. [K] / [CO1]
   - Weight Management & Metabolic Health (1 mark)
   - Cardiovascular/ Respiratory health/ Musculoskeletal Strength (1 mark)
5. Mention 2 consequences of low carbohydrate availability during sports. [U] / [CO3]

- Early fatigue – Muscles run out of glycogen, leading to tiredness and reduced endurance. (1 mark)
- Decreased performance – The athlete's speed, strength, and concentration drop because the body lacks its main energy source. (1 mark)

6. What is meant by physical fitness? [K] / [CO1]
   - Physical fitness refers to the state of being physically healthy and fit. (1 mark)
   - It typically includes components such as cardiovascular endurance, muscular strength, muscular endurance, flexibility, and body composition. (1 mark)

7. State 2 early symptoms of dehydration. [U] / [CO2]
   - Thirst/ dry mouth (1 mark)
   - Fatigue (1 mark)

8. State 2 effects of excessive fat intake on exercise performance. [U] / [CO3]
   - Slower digestion and discomfort – High fat intake can cause stomach upset, bloating, or nausea during exercise. (1 mark)
   - Reduced energy availability – Fat is a slow energy source, so relying too much on it can lower endurance and delay energy release needed for performance. (1 mark)

9. Write any 2 consequences of using narcotics in sports. [U]/ [CO4]
   - Reduced alertness and coordination (1 mark)
   - Addiction and health problem (1 mark)

10. Write 2 methods used to measure energy expenditure. [U] / [CO3]
    (1 mark each for any of the two methods mentioned below)
    - Direct calorimetry
    - Indirect calorimetry
    - Doubly labelled water
    - Accelerometers and Heart Rate Monitors

11. State 2 nutrients important for power events. [K] / [CO1]
    - Carbohydrates (CHO) - Critical for high-intensity efforts and glycogen replenishment. (1 mark)
    - Protein (Muscle Repair & Adaptation) - Supports muscle recovery, reduces soreness, and promotes adaptation. (1 mark)

12. Mention 2 effects of inadequate protein intake in athletes. [U] / [CO2]
    - Impaired Muscle Repair -can lead to prolonged soreness and decreased performance. (1 mark)
    - Decreased Immune Function. (1 mark)

(2x10 = 20)

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. Discuss the impact of sweat rate on fluid replacement strategies. [U] / [CO3]
    - **Fluid replacement planning:** Knowing an athlete's sweat rate helps estimate how much fluid needs to be consumed before, during, and after exercise to prevent dehydration or overhydration**.** (2 marks)
    - **Performance impact:** If fluid loss isn't replaced according to sweat rate, dehydration can reduce endurance, increase heart rate and body temperature, and impair performance. (2 marks)

14. Discuss the role of nutritional ergogenic aids in enhancing performance. [An] / [CO3]
    - **Nutritional ergogenic** aids are nutrients or supplements that enhance athletic performance by improving energy availability, strength, and endurance. (2 marks)
    - Examples like caffeine, creatine, and carbohydrate drinks help delay fatigue and promote faster recovery after exercise. (2 marks)

15. Differentiate between team sports and individual sports with examples. [K] / [CO1]
    - **Team sports** involve organized groups of players working together to achieve a common goal. Eg) soccer, volleyball, basketball. (2 marks)
    - **Individual sports**, on the other hand, focus on a single athlete competing against others. Success relies on personal skill, determination, and self-discipline. Eg) swimming, gymnastics (2 marks)

16. Discuss the importance of electrolytes in maintaining muscle and nerve function during exercise. [U] / [CO2]
    - **Muscle Function** -Electrolytes transmit nerve impulses that trigger muscle contractions. Imbalances may cause cramps, spasms, fatigue, or muscle weakness. (2 marks)
    - **Nerve Transmission** - Sodium and potassium gradients are vital for nerve signal conduction, directly affecting coordination and reflexes during physical activity. (2 marks)

17. Discuss the importance of calcium for bone health and muscle contraction in athletes [U] / [CO2]
    - **Muscle Contraction** - During exercise, calcium ions are released inside muscle cells, triggering contractions that make movement possible. (2 marks)

- **Role in Bone Health -** Adequate calcium intake helps maintain bone density and prevent osteoporosis or fractures in athletes. (2 marks)

18. Discuss the importance of antioxidants in reducing exercise – induced muscle damage. [U] / [CO3]
    - During intense exercise, the body produces free radicals (reactive oxygen species) that can damage muscle cells. (2 marks)
    - Antioxidants such as vitamins C, E, beta-carotene, and compounds from fruits and vegetables help neutralize these free radicals, preventing cell damage. (2 marks)

19. State 4 examples of mechanical aids with their benefits. [U] / [CO4]
    - **High-tech footwear** – Improves running efficiency (½ mark)
    - **Compression garments** – Enhance blood flow (½ mark)
    - **Wrist straps** – Improve grip strength and support during weightlifting. (½ mark)
    - **Running shoes** – Provide cushioning and reduce impact injuries. (½ mark)

(4x5=20)

Part C

Essay Type Questions
Answer any **1** questions
Each question carries **10** marks

20. Explain the objective and functions of NADA in controlling doping practices in sports. [U] / [CO4]

Objective (2marks)

- The main objective of the National Anti-Doping Agency (NADA) is to promote fair play and integrity in sports by ensuring that all athletes compete in a drug-free environment.
- It aims to prevent and control doping practices in India in line with the World Anti-Doping Code (WADA)**.**

Functions (8 marks)

- **Testing and Monitoring:** Conducts dope tests on athletes during and outside competitions to detect prohibited substances or methods.
- **Education and Awareness:** Educate athletes and coaches about the dangers and consequences of doping.
- **Result Management and Sanctions:** Handles the results of doping tests and imposes penalties or suspensions on athletes found guilty of violations.

- **Coordination with WADA and Sports Bodies:** Works with the **World Anti-Doping Agency (WADA)** and national sports federations to ensure compliance with anti-doping regulations.

21. Discuss the role and importance of post event recovery meals in sports performance.

[K] / [CO1]

Importance of Post-Event Meals (2 marks)

- After intense physical exertion, the body enters a catabolic state where glycogen stores are depleted, muscle tissue is damaged, and fluids along with electrolytes are lost. (1 mark)
- Consuming a post-event meal is crucial for recovery, helping to restore balance and initiate the healing process. (1 mark)

Key Roles of Post-Event Meals (8 marks)

- **Replenishing Glycogen Stores -** Carbohydrates are essential to refill glycogen levels that are diminished during exercise. Simple carbs, like fruits, are effective immediately after an event.
- **Repairing Muscle Tissue-** Protein intake is vital to promote muscle recovery.
- **Restoring Hydration -** Rehydration is critical. Replace lost fluids with water and electrolyte solutions to support recovery.
- **Supporting Immune Function -** Adequate nutrition helps bolster the immune system, which can be compromised after intense physical activity.

(1x10=10)